Speaker Information

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Talk Information

Large Volume Electron Microscopy Reveals Synaptic Architecture of Cortical Circuits in the Primate Prefrontal Cortex

Over the past decade, large-volume electron microscopy (vEM) has enabled detailed analysis of cortical synaptic wiring. Here, we developed an efficient vEM pipeline to study synaptic microcircuitry in the prefrontal cortex (PFC) of the common marmoset. We optimized an automated tape-collecting ultramicrotome (ATUM) to stably collect over 1,000 serial ultrathin sections (50 nm), which were imaged at high resolution (2.38 nm/pixel) using a high-throughput TEM system (Blade). This enabled acquisition of ultrastructural data across a 1.7 × 1.2 mm cortical area within about one month. To examine cortico-cortical connectivity, we co-injected anterograde (GFP) and retrograde (td-Tomato) AAV tracers into area 10, labeling projections between areas 9 and 10. Confocal imaging identified labeled structures, followed by vEM analysis of the same regions. Anterogradely labeled axons primarily targeted dendritic spines of pyramidal neurons (89%), with fewer synapses on non-pyramidal cells (11%). In contrast, retrogradely labeled pyramidal neurons mainly innervated other pyramidal cells (65%), with additional connections to fast-spiking basket cells (32%) and rare contacts with double bouquet cells (2%). These results provide new insights into primate PFC circuitry and demonstrate the power of vEM for resolving synaptic connectivity at nanometer resolution.

Research Overview (for collaboration discussion)

[Research Interests] Connectome in cortex

[Current Research Projects]

1) Marmoset cortical microcircuit investigation using large volume EM

2) EM analysis of astrocytic processes approaching to synapse in cortex

[Techniques] large volume EM, CLEM, in vivo imaging, EM image analysis

[Aspects or techniques of interest for future collaboration]

Short Bio

I would like to find out the microcircuit architecture of neocortex. My main tools are EM and in vivo imaging.